

Bode Plot

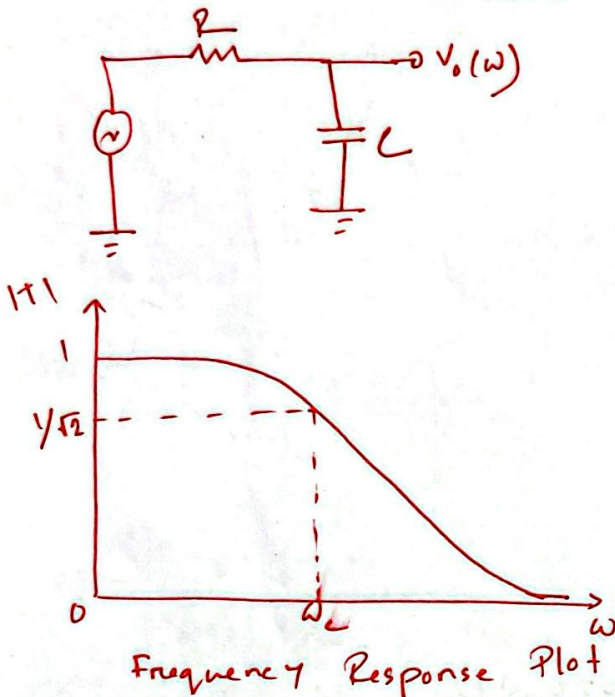
①

It A technique to draw the frequency response plot.

Transmission magnitude is expressed in decibel and frequency axis is taken in log scale.

plot of $20 \log |T|$ (in dB)
vs frequency (in log scale)

Passive RC LP filter:



$$T(s) = \frac{V_o}{V_i} = \frac{1}{1 + s/\omega_c} \quad \omega_c = \frac{1}{RC}$$

$$T(j\omega) = \frac{V_o}{V_i} = \frac{1}{1 + j\omega CR}$$

$$|T(j\omega)| = \left| \frac{1}{1 + j\omega CR} \right|$$

$$= \frac{1}{|1 + j\omega CR|}$$

$$= \frac{1}{\sqrt{1 + (\omega CR)^2}}$$

Frequency Response Plot

Gain Magnitude $|T|$ vs Frequency.

$$= 1, \quad \omega = 0$$

$$20 \log |T| \text{ (dB)} = 0, \quad \omega = \infty$$

$$20 \log |T(s)|$$

$$= 20 \log \frac{1}{|1 + s/\omega_c|}$$

$$20 \log |T(j\omega)|$$

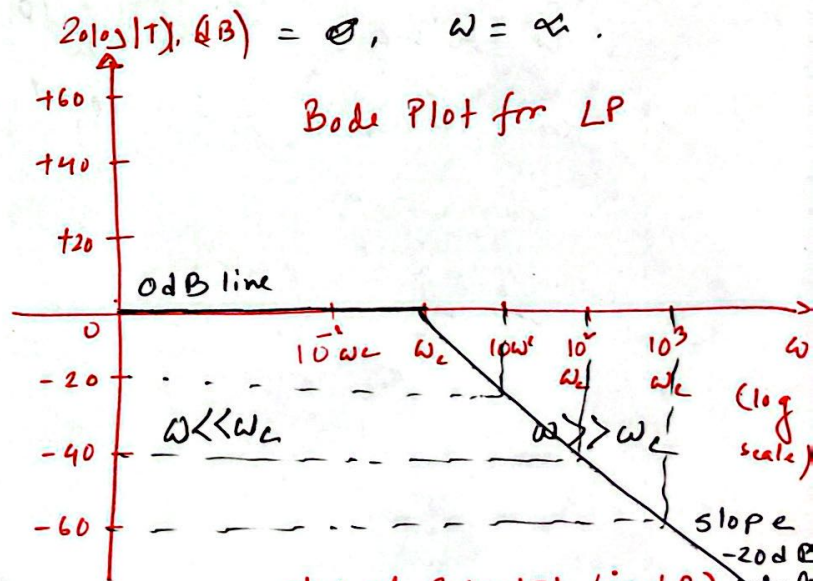
$$= 20 \log \left| \frac{1}{1 + j\omega/\omega_c} \right|$$

$$= 20 \log \frac{1}{\sqrt{1 + (\omega/\omega_c)^2}}$$

$$= 20 \log [1 + (\omega/\omega_c)^2]^{-1/2}$$

$$= -10 \log [1 + (\omega/\omega_c)^2]$$

Bode Plot for LP



plot of $20 \log |T|$ (in dB)
vs frequency (in log scale)

Evaluate for two ranges of ω

① $\omega \ll \omega_c$

② $\omega \gg \omega_c$

① $\omega \ll \omega_c$

$$\therefore \frac{\omega}{\omega_c} \ll 1$$

$$\begin{aligned}\therefore 20 \log |T(j\omega)| &= -10 \log \left[1 + (\omega/\omega_c)^2 \right] \\ &= -10 \log 1 \\ &= 0 \text{ dB}\end{aligned}$$

$$\therefore \text{For, } \omega \leq \frac{\omega_c}{10}, \omega \ll \omega_c$$

② $\omega \gg \omega_c$

$$\omega/\omega_c \gg 1$$

$$20 \log |T(j\omega)| = -20 \log (\omega/\omega_c)$$

$$\text{for, } \omega = 10 \omega_c = -20 \log (10) = -20 \text{ dB}$$

$$\omega = 10^2 \omega_c = -20 \log (10^2) = -40 \text{ dB}$$

$$\omega = 10^3 \omega_c = -20 \log (10^3) = -60 \text{ dB}$$

As a summary,

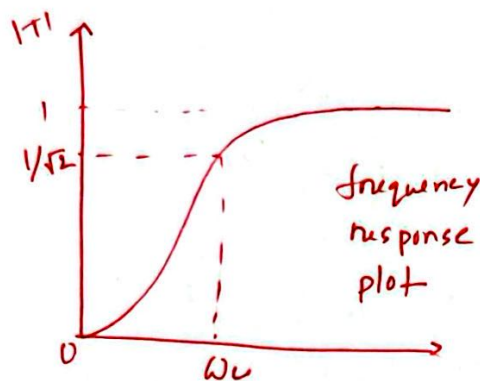
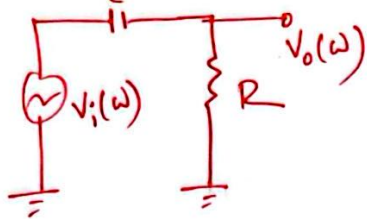
$$T(s) = \frac{V_o}{V_i} = \frac{1}{1 + s/\omega_c}$$

$$20 \log |T(s)| = 20 \log \frac{1}{|1 + s/\omega_c|}$$

① $\omega < \omega_c$, 0 dB line, valid for $\omega \leq \frac{\omega_c}{10}$

② $\omega > \omega_c$, slanted line with slope -20 dB/decade
valid for $\omega \geq 10 \omega_c$.

HP filter:



$$T(s) = \frac{V_o}{V_i} = \frac{1}{1 + \omega_c/s} \quad \omega_c = \frac{1}{RC}$$

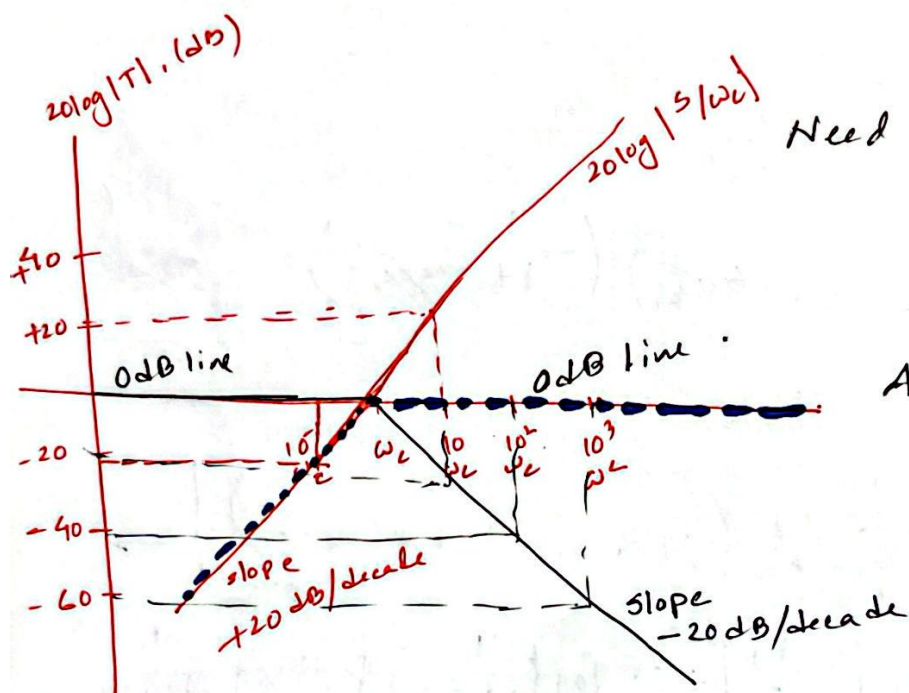
$$= \frac{1}{1 + \omega_c/s} \times \frac{s/\omega_c}{s/\omega_c}$$

$$= \frac{s/\omega_c}{1 + s/\omega_c}$$

$$20 \log |T(s)| = 20 \log \frac{|s/\omega_c|}{|1 + s/\omega_c|}$$

$$= 20 \log |s/\omega_c| + 20 \log \frac{1}{|1 + s/\omega_c|}$$

same as LP filter.



Need to plot,

$$20 \log |s/\omega_c| = 20 \log \left| \frac{j\omega}{\omega_c} \right|$$

$$= 20 \log \left(\frac{\omega}{\omega_c} \right)$$

$$\text{At } \omega = \omega_c$$

$$= 20 \log 1$$

$$= 0 \text{ dB}$$

$$\omega = 10 \omega_c$$

$$= 20 \log (10)$$

$$= 20 \text{ dB}$$

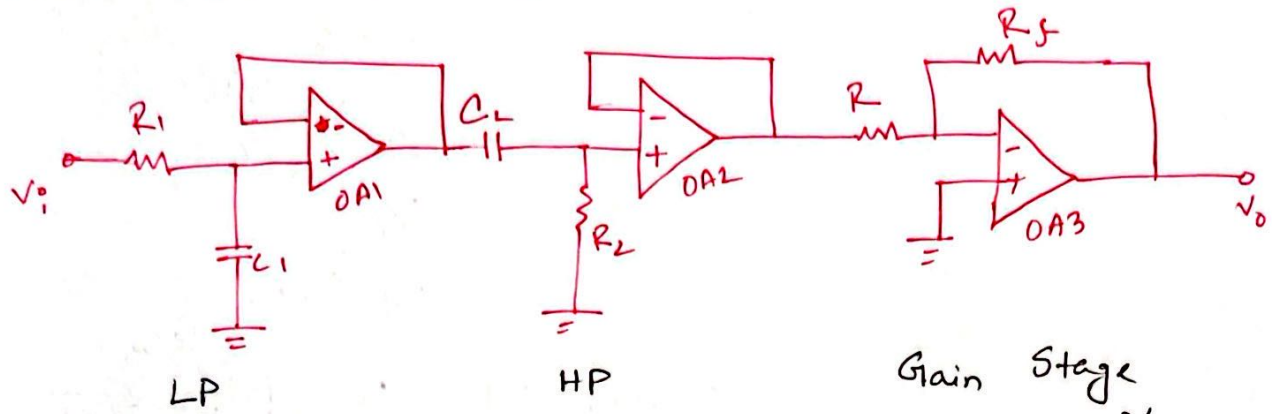
$$\omega = 10^{-1} \omega_c$$

$$= 20 \log (10^{-1})$$

$$= -20 \text{ dB}$$

$$20 \log |T(s)| = 20 \log \left| \frac{s}{\omega_c} \right| + 20 \log \left| \frac{1}{1 + s/\omega_c} \right|$$

Bode Plot: Band Pass filter;



$$T_1(s) = \frac{V_{o1}}{V_i} = \frac{1}{1+s/\omega_{c1}}$$

$$T_2(s) = \frac{V_{o2}}{V_{o1}} = \frac{1}{1+\omega_{c2}/s}$$

$$T_3(s) = \frac{V_o}{V_{o2}} = -\frac{R_f}{R}$$

∴ Overall transfer function,

$$T(s) = \frac{V_o}{V_i} = \frac{V_o}{V_{o2}} \times \frac{V_{o2}}{V_{o1}} \times \frac{V_{o1}}{V_i} \\ = -\frac{R_f}{R} \left(\frac{1}{1+s/\omega_{c1}} \right) \left(\frac{1}{1+\omega_{c2}/s} \right)$$

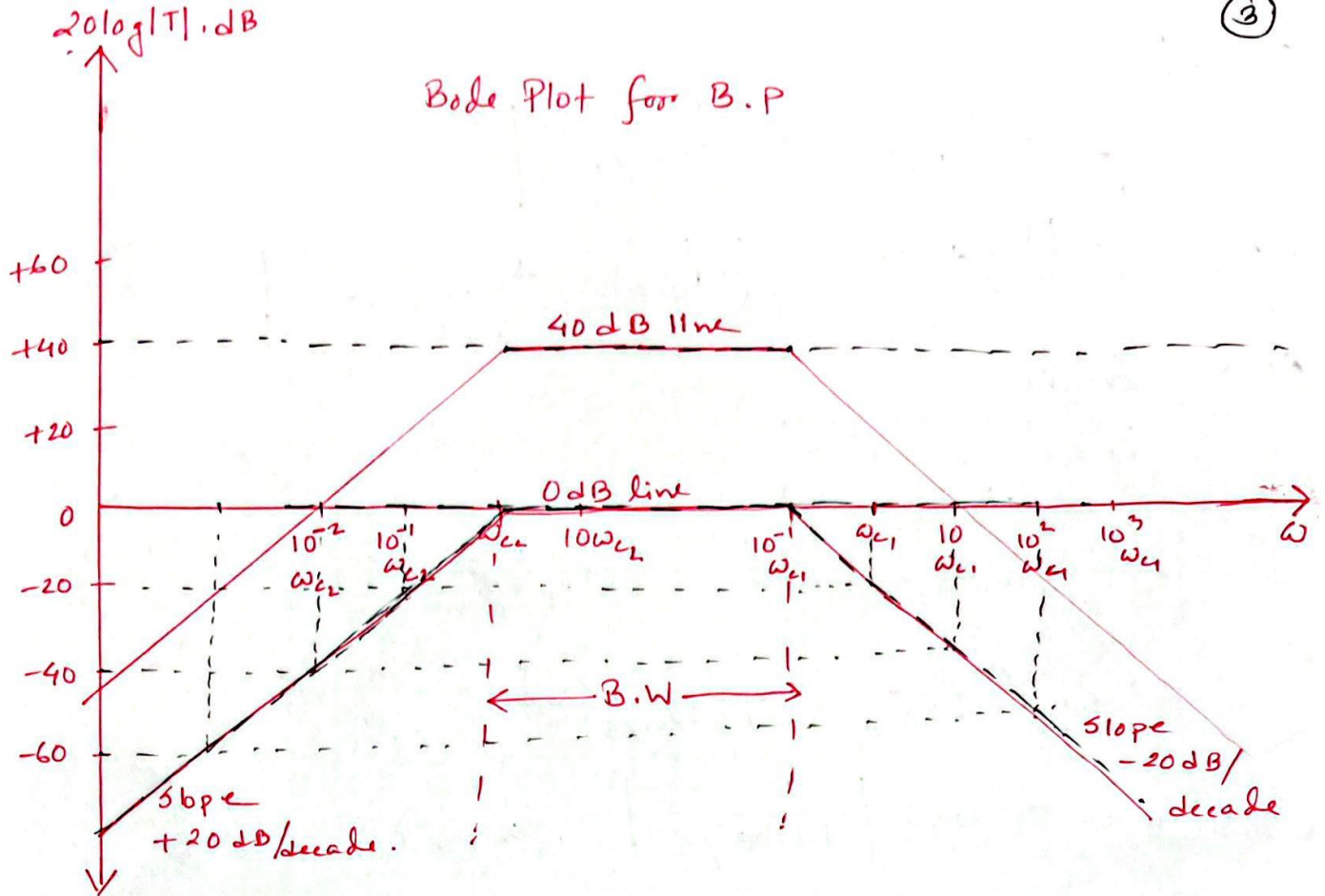
$$\boxed{\omega_{c1} > \omega_{c2}}$$

$$20 \log |T(s)| = 20 \log \left| -\frac{R_f}{R} \left(\frac{1}{1+s/\omega_{c1}} \right) \left(\frac{1}{1+\omega_{c2}/s} \right) \right| \\ = 20 \log \left| -\frac{R_f}{R} \right| + 20 \log \left| \frac{1}{1+s/\omega_{c1}} \right| + 20 \log \left| \frac{1}{1+\omega_{c2}/s} \right|$$

$$20 \log \left| -\frac{R_f}{R} \right| = 20 \log \left(\frac{R_f}{R} \right)$$

$$= 20 \log (100)$$

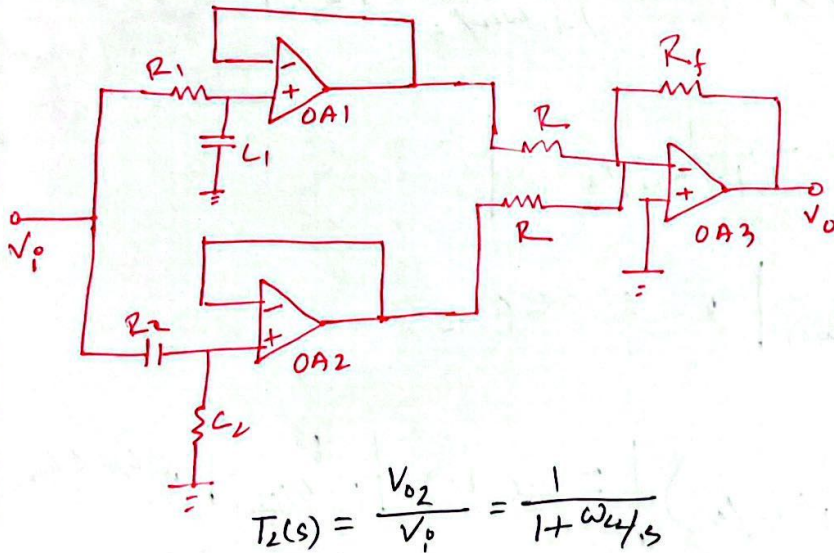
$$= 40 \text{ dB}$$



Band Stop Filter:

$$T_1(s) = \frac{V_{o1}}{V_i} = \frac{1}{1 + s/\omega_{c1}}$$

$$\boxed{\omega_{c2} > \omega_{c1}}$$



$$V_o = -\frac{R_f}{R} (V_{o1} + V_{o2})$$

$$T(s) = \frac{V_o}{V_i} = -\frac{R_f}{R} \left(\frac{V_{o1}}{V_i} + \frac{V_{o2}}{V_i} \right)$$

$$= -\frac{R_f}{R} (T_1(s) + T_2(s))$$

$$= -\frac{R_f}{R} \left(\frac{1}{1 + s/\omega_{c1}} + \frac{1}{1 + s/\omega_{c2}} \right)$$

∴ Overall transfer function:

$$T(s) = \frac{V_o}{V_i} = -\frac{R_f}{R} \left(\frac{1}{1 + s/\omega_{c1}} + \frac{1}{1 + s/\omega_{c2}} \right)$$

$$\begin{aligned}
 & 20 \log |T(s)| \\
 &= 20 \log \left| -\frac{R_f}{R} \left(\frac{1}{1+s/\omega_{c1}} + \frac{1}{1+\omega_{c2}/s} \right) \right| \\
 &= 20 \log \left| -\frac{R_f}{R} \right| + 20 \log \left| \left(\frac{1}{1+s/\omega_{c1}} + \frac{1}{1+\omega_{c2}/s} \right) \right|
 \end{aligned}$$

At high freq end. $\omega \gg \omega_{c1}$

$$\Rightarrow \frac{\omega}{\omega_{c1}} \gg 1$$

$$\Rightarrow \frac{s}{\omega_{c1}} \gg 1, \quad \frac{1}{1+s/\omega_{c1}} \ll 1$$

$$\begin{aligned}
 \therefore 20 \log \left| \left(\frac{1}{1+s/\omega_{c1}} + \frac{1}{1+\omega_{c2}/s} \right) \right| \\
 = 20 \log \left| \frac{1}{1+\omega_{c2}/s} \right|, \quad \omega \gg \omega_{c1}
 \end{aligned}$$

At low freq end,

$$\omega \ll \omega_{c2}$$

$$\therefore \frac{\omega_{c2}}{\omega} \gg 1$$

$$\Rightarrow \frac{\omega_{c2}}{s} \gg 1, \quad \frac{1}{1+\omega_{c2}/s} \ll 1$$

$$\begin{aligned}
 \therefore 20 \log \left| \left(\frac{1}{1+s/\omega_{c1}} + \frac{1}{1+\omega_{c2}/s} \right) \right| \\
 = 20 \log \left| \frac{1}{1+s/\omega_{c1}} \right|, \quad \omega \ll \omega_{c2}
 \end{aligned}$$

$$\begin{aligned}
 \therefore 20 \log \left| \left(\frac{1}{1+s/\omega_{c1}} + \frac{1}{1+\omega_{c2}/s} \right) \right| &\approx 20 \log \left| \frac{1}{1+s/\omega_{c1}} \right|, \quad \omega \ll \omega_{c2} \\
 &\approx 20 \log \left| \frac{1}{1+\omega_{c2}/s} \right|, \quad \omega \gg \omega_{c1}
 \end{aligned}$$

$$\omega_{c2} > \omega_{c1}$$

